

Section A

Answer **all** the questions in this section in the spaces provided.

- 1** The mass spectrometer made it possible to determine the masses of atoms very accurately, and it became clear that not all atoms of the same element had the same mass.

The accurate masses gave rise to the idea of the isotope:

“Isotopes have the same atomic number, but different mass numbers.”

- (a) (i)** Give the symbols showing the nucleon numbers and charges of the following three particles, **K**, **L** and **M**.

You may refer to the example given for the particle, **A**

Particle	Protons	Neutrons	Electrons	Symbol
K	6	8	6	
L	7	7	10	
M	8	7	7	
A	5	8	5	$^{13}_5\text{A}$

[3]

- (ii)** Use the Data Booklet, identify which are the usual isotopes of the element concerned.

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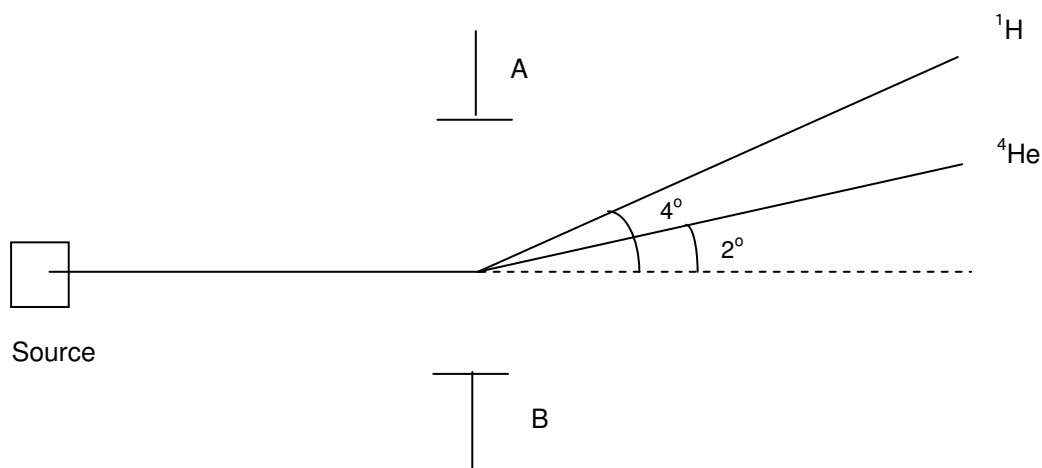
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[3]

- (b) A plasma is a gaseous mixture in which the atoms have been completely stripped of their electrons, leaving bare nuclei. Because of possible use in controlled nuclear fusion reactions, plasma behaviour has been extensively studied. When passed between two plates carrying a certain electric charge, ^1H and ^4He nuclei are deflected as follows.

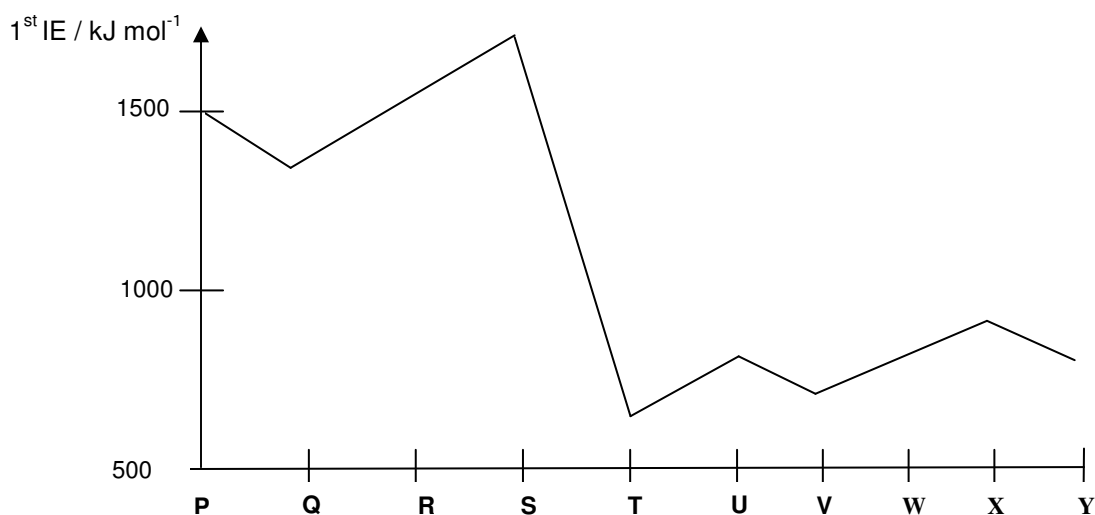


Giving reason for your answer, suggest the polarity (+ or -) of plate A.

[2]

- (c) In an analytical chemistry laboratory, the plasma torch is often used to ionise a variety of samples during their analyses.

In one series of analyses, samples of unknown elements **P – Y** were passed through the plasma torch and their first ionisation energies were recorded and plotted as shown.



Elements **P – Y** have atomic number below 20.

- (i) Identify the Group in the Periodic Table to which element **V** belongs and write its ground state electronic configuration.

Element **V** belongs to Group

Electronic configuration of **V**:

[2]

- (ii) Element **X** forms more than one stable chloride. State the formula of one of these chlorides. Draw the dot-and-cross diagram of the chloride of **X** that you have stated and predict its shape.

Formula of the chloride:

Dot-and-cross diagram:

Shape of chloride of **X**:

[3]

- (iii) A pair of elements exists among the elements **P – Y** belonging to the same group. Identify them, stating the group to which they belong.

Pair: Group: [2]

[Total: 15m]

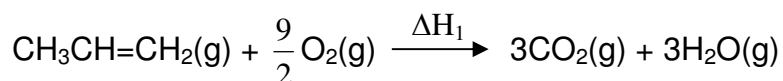
- 2** Surplus alkenes can be used as fuels as well as starting materials for petrochemical processes. Being hydrocarbons, alkenes can burn, transferring energy to their surroundings.

- (a) What do you understand by the term *bond energy*?

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[1]

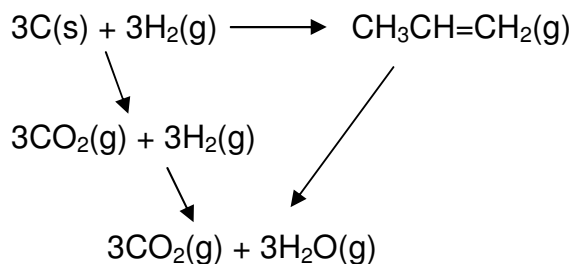
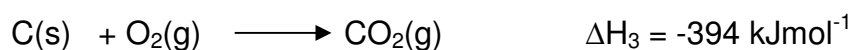
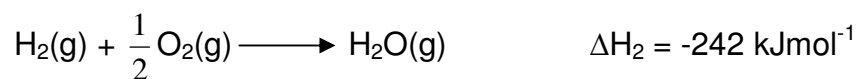
- (b) The equation for combustion of propene, $\text{CH}_3\text{CH}=\text{CH}_2$, is



By using appropriate bond energy data given in the *Data Booklet*, calculate a value for the enthalpy change of the above reaction, ΔH_1 , involving propene.

[3]

- (c) Using the answer obtained in (b) and with the following enthalpy changes for the reactions given below, use the energy cycle provided to calculate the enthalpy change of formation, ΔH_f , of propene.



[2]

- (d) A sample of propene was ignited in a calorimeter to heat up 150 cm^3 of water. The temperature of the water rose by 45°C .

Calculate the mass of propene combusted.

Assume the specific heat capacity of water to be $4.2 \text{ Jg}^{-1}\text{K}^{-1}$.

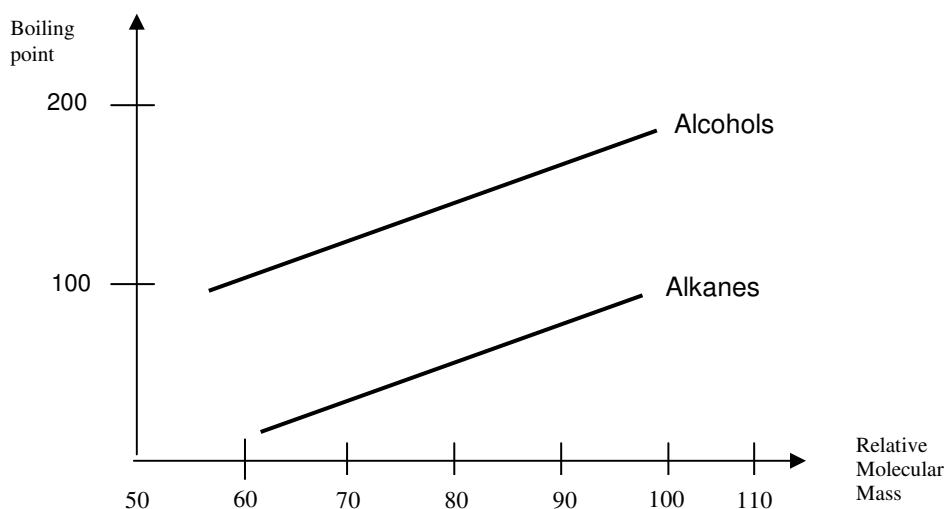
[2]

- (e) The growing demand for gasoline and the potential shortage of oil is one reason alcohol fuel is seen as a possible replacement. This technically feasible fuel is economically competitive if oil supplies should run short. Although it is more expensive than gasoline, alcohol as fuel is still possible in blends with gasoline.

Every chemist knows alcohol and gasoline will mix and every engineer knows this mixture will drive an internal combustion engine.

Alkanes and alcohols are highly combustible.

The general trend of boiling points of alkanes and alcohols against relative molecular mass is shown in the graph below.



- (i) Explain, in terms of the bonding present, the trends shown by the above graph.

[3]

- (ii) On the above graph, indicate the trend of boiling point of carboxylic acids against relative molecular mass.

[1]

- (f) The table below shows the sources and the enthalpy change of combustion of two common fuels used in vehicles.

Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}
Octane (C_8H_{18})	Refined from crude oil	-5460.6
Ethanol ($\text{C}_2\text{H}_5\text{OH}$)	Hydration of ethene Fermentation	-1366.2

- (i) Define, using ethanol as an example, the term *enthalpy change of combustion* with an appropriate equation.

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[1]

- (ii) The fuel value (in kJ g^{-1}) of a substance is the heat energy *released* when 1 g of the substance is combusted.
Calculate the fuel value for each of the two fuels.

[1]

- (iii) With other information, and your answer in (f), compare the relative advantages and disadvantages of using octane and ethanol as fuels for vehicles.

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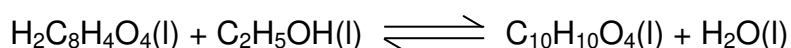
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[3]

[Total: 17m]

- 3 Upon heating, phthalic acid, $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$, reacts with ethanol according to the equation as shown below.

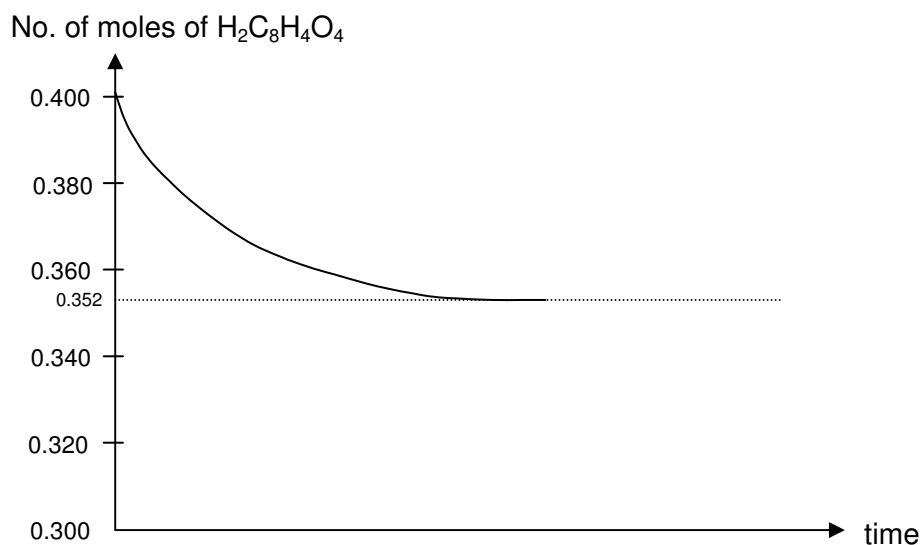


The equation does not really indicate the problems with the reaction. To get a good yield of the product, $\text{C}_{10}\text{H}_{10}\text{O}_4$, the reaction has to be managed carefully since it is reversible.

A mixture with the following composition was heated in a closed container and the reaction allowed to reach equilibrium.

Compound	No. of moles
$\text{H}_2\text{C}_8\text{H}_4\text{O}_4$	0.400
$\text{C}_2\text{H}_5\text{OH}$	0.600
$\text{C}_{10}\text{H}_{10}\text{O}_4$	0.100
H_2O	0.100

The graph shows how the number of moles of $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$ varies with time.



- (a) To analyse the composition of the reaction mixture, a fixed volume was extracted at regular time intervals and titrated **rapidly** against a standard solution of alkali. Explain why the titration has to be carried out rapidly.

[1]

- (b) Write an expression for the equilibrium constant, K_c , for the reaction above.

[1]

- (c) What is the value of the equilibrium constant K_c for this reaction. Show how you derive at your answer.

[2]

- (d) Explain how the equilibrium composition would change if cold concentrated sulphuric acid was added to the equilibrium mixture.

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[2]

- (e) The reaction was repeated at the same temperature with 0.300 moles each of $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$ and $\text{C}_2\text{H}_5\text{OH}$ and 0.200 moles each of $\text{C}_{10}\text{H}_{10}\text{O}_4$ and H_2O . Determine the concentration of $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$ at this equilibrium, in a 1 dm^3 container.

[2]

[Total: 8m]

Section B

Answer **two** of the three questions in this section on separate paper.

- 4** Halogenoalkanes are saturated compounds. This limits the reaction available, since saturated compounds cannot undergo addition reactions. Halogenoalkanes undergo two major types of reaction, substitution and elimination.

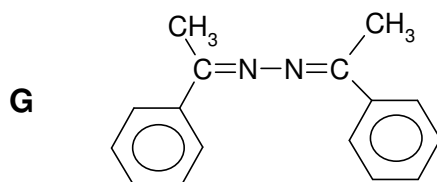
Two isomers **A** and **B** have molecular formula $C_2H_4Br_2$.

- (a) On hydrolysis, **A** yields an aldehyde **C** but **B** yields a diol **D**. Draw structural formulae for **A**, **B**, **C** and **D**. [4]

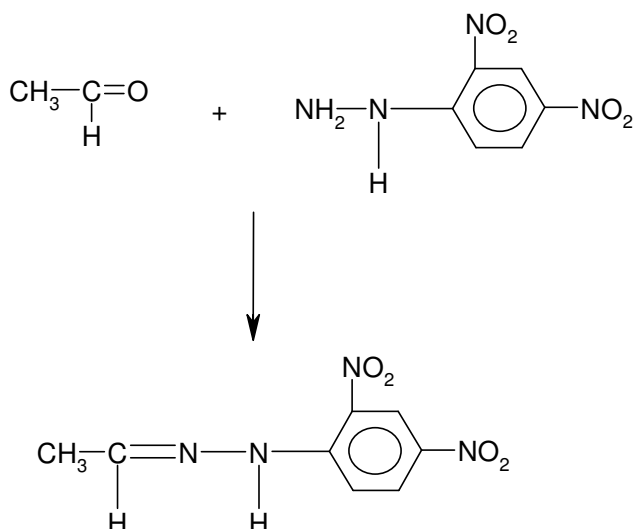
- (b) **D** may be oxidised to a dicarboxylic acid **E**.

Draw the full structural formula of the cyclic ester **F** obtained when one molecule of **D** reacts completely with one molecule of **E**. On your diagram, mark the bond angles at the carbon atoms in the structural formula of **F**. [4]

- (c) The compound **C** (an aldehyde) is a carbonyl compound. This carbonyl functional group is found in many important biological molecules, from insect pheromones to human sex hormone. This functional group is easily identified by 2,4-dinitrophenylhydrazine.



Compound **G** can be synthesised from hydrazine, N_2H_4 . The reaction is similar to that undergone between a carbonyl compound, such as ethanal, and 2,4-dinitrophenylhydrazine.



- (i) Identify the carbonyl compound that is needed in the synthesis of **G**.
Write a balanced equation for the reaction and state the type of reaction undergone. [4]
- (ii) Is compound **G** planar? Explain. [3]
- (iii) State the type of isomerism displayed by **G** and draw two possible isomers to illustrate. Label the isomers. [5]

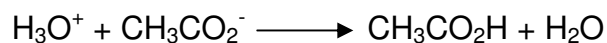
[Total: 20m]

- 5 (a) Malt vinegar is made by malting barley, which causes the starch in the grain to first turn into sugar. An ale is then brewed from the sugar and allowed to turn into vinegar, which is then aged. Malt vinegar typically contains 5% by volume of ethanoic acid, CH_3COOH .
(Density of CH_3COOH is 1.05 g cm^{-3} , K_a of CH_3COOH is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$)
Calculate the pH of a 100 cm^3 solution of malt vinegar. [4]

Thymol blue is an indicator which has two working ranges as shown in the table below.

pH	Color of thymol blue
< 1.2	Red
2.6 – 8.0	Yellow
> 9.5	Blue

When two drops of thymol blue are added to 100 cm³ of 0.10 mol dm⁻³ hydrochloric acid at 25°C, a red solution is obtained. The following reaction occurs when sodium ethanoate, CH₃CO₂Na (Mr = 82), is added to the solution:



- (b) When 1.64 g of solid sodium ethanoate is added to the hydrochloric acid solution, the solution turns yellow.
- Calculate the number of moles of ethanoic acid formed.
 - Explain, qualitatively, why the solution turns yellow.
 - Name the resulting mixture obtained and hence calculate the pH of the resulting mixture.

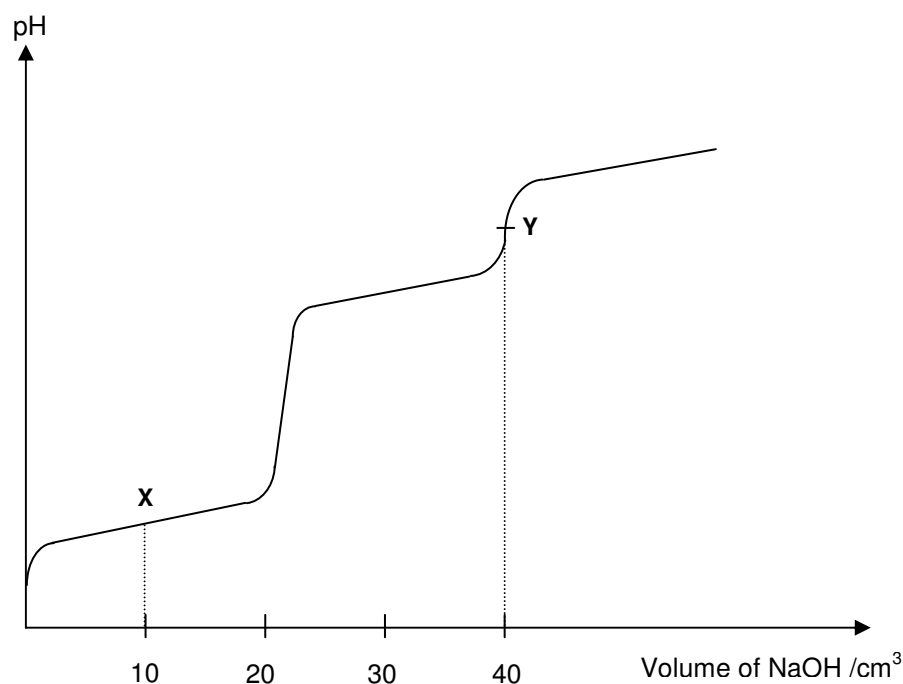
[8]

- (c) Vitamin C contains ascorbic acid. Ascorbic acid is a dibasic acid with the following dissociation constants.

$$K_{a1} = 8.0 \times 10^{-5} \text{ mol dm}^{-3}$$

$$K_{a2} = 1.6 \times 10^{-11} \text{ mol dm}^{-3}$$

0.10 mol dm⁻³ NaOH solution is added slowly to 20.0 cm³ of 0.10 mol dm⁻³ of ascorbic acid and the change in pH is followed and shown below:



- (i) Calculate the initial pH of a 0.100 mol dm⁻³ solution of ascorbic acid, stating any assumptions you make.
(Use the formula of H₂A as ascorbic acid.)
- (ii) What is the pH at point X. Explain your answer.
- (iii) This titration employs Bromocresol green as the indicator.

The following table shows the change of color of Bromocresol green indicator.

[H ⁺] / mol dm ⁻³	10 ⁻¹	10 ⁻²	10 ⁻³	10 ⁻⁴	10 ⁻⁵	10 ⁻⁶	10 ⁻⁷
Color of bromocresol green	Yellow	Yellow	Yellow	Green	Blue	Blue	Blue

What is the color of Bromocresol green indicator in the ascorbic acid solution before the titration?

State whether Bromocresol green is a suitable indicator for use in titration of ascorbic acid with sodium hydroxide. [8]

[Total: 20m]

- 6 There is a wide variety of oxidation states in the transition metals. Manganese, Mn, provides many examples on this property.

Nuclear medicine makes wide use of a metastable form of an isotope of technetium $^{99m}_{43}\text{Tc}$. $^{99m}_{43}\text{Tc}$ is injected into the blood stream as the compound NaTcO_4 , to monitor blood flow in some vital internal organs. Technetium and manganese are in the same group of the Periodic Table.

- (a) What is the oxidation state of Tc in NaTcO_4 ? [2]
- (b) An acidified solution of NaTcO_4 may react with iodide ions to give iodine, and itself is reduced to Tc^{2+} . Derive an equation to represent this reaction.
(You are to include all essential half-equations.) [3]
- (c) Technetium in the ion TcO_4^- has no unshared electrons. Predict the shape of TcO_4^- . [2]

The Winkler method can be used to determine the amount of dissolved oxygen in a water sample. In an experiment, manganese(II) sulphate solution is first added to a 100 cm^3 water sample in a conical flask, followed by alkaline saturated potassium iodide solution in excess.

A brown precipitate of manganese(III) hydroxide is formed. Concentrated sulphuric acid is then added and the solution is carefully mixed until the precipitate disappears, leaving a dark brown solution.

This solution requires 7.50 cm^3 of $0.0125\text{ mol dm}^{-3}$ sodium thiosulphate for titration.

- (d) Write a balanced ionic equation for the reaction of manganese(II) sulphate and oxygen in the water sample under alkaline condition. [2]

- (e) Write an equation for the disappearance of the brown precipitate on adding concentrated sulphuric acid. Explain why the product appears as a dark brown solution. **[4]**
- (f) Write an equation for the titration reaction and hence calculate the concentration of the dissolved oxygen in the water sample in mg dm^{-3} . **[7]**

[Total: 20m]